Recovering and Redirecting Peptide Backbone Conformation via Contact-Conditioned Cα Generation in a Cross-Reactive pMHC–TCR System

Sergio E. Mares[1,2]
[1]Adimab, Mountain View, CA 94043, USA
[2]Center for Computational Biology, University of California, Berkeley, CA 94720, USA

Abstract

Cross-reactive TCR–pMHC systems, in which one receptor engages chemically distinct peptides through distinct bound backbone conformations, raise a question orthogonal to conventional epitope design: can a structure-conditioned generative model be steered, from receptor-side information alone, toward a specific target backbone conformation — including, deliberately, one other than the conformation nearest its conditioning input? Using a minimal two-state system — the DMF5 TCR bound to HLA-A*02:01, which accommodates two different decameric peptides (GIG and DRG) via backbones separated by 2.87Å Cα RMSD — we ask whether RFdiffusion, conditioned only on receptor-side contact residues and given no peptide-backbone template, can generate a Cα backbone that recovers a peptide's own native conformation or crosses into the alternate one. We score de novo backbones against a three-criterion test calibrated to native molecular-dynamics ensembles rather than to a single crystal structure, and address a recurring evaluation pitfall by correcting for a backbone-threading artifact that raw RMSD cannot detect. We find that both recovery and redirection are achievable but rare, gated by the breadth of conditioning rather than by any specific curated residue set, and that once that breadth is met their frequency is governed by sampling depth. A templating control further shows that directly supplying backbone geometry constrains the target conformation far more tightly than contact conditioning alone, locating register in a spatially confined signal that receptor-side contacts only weakly determine. These results, from a single system, map the current capability boundary of contact-conditioned backbone generation and separate what it can retrieve from what must still be supplied as explicit geometry.

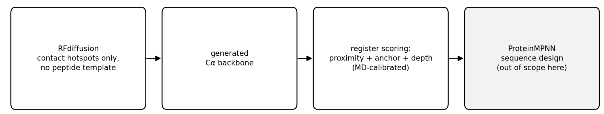

Figure 1: **Pipeline schematic.** RFdiffusion (contact-hotspot conditioning, contig `10-10`, no peptide template) $\rightarrow$ generated Cα backbone $\rightarrow$ three-criterion register scoring against MD-calibrated thresholds $\rightarrow$ (downstream, out of scope here) ProteinMPNN sequence design.

1 Introduction

Cross-reactive TCR–pMHC pairs that engage chemically distinct peptides via distinct bound backbone conformations pose a design question orthogonal to conventional epitope design: can a structure-generative model be conditioned, using only information available on the receptor side, to produce a specific target backbone conformation — including, deliberately, the conformation other than the one nearest to the conditioning input? We address this question exclusively at the Cα backbone level. RFdiffusion generates backbone coordinates without an associated sequence at this stage of the pipeline; consequently every metric defined here operates on Cα geometry, treated throughout as a geometric rather than sequence property. GIG and DRG are not the same peptide sequence adopting two registers of a shared groove placement — they are two different decamers (SMLGIGIVPV and MMWDRGLGMM) that DMF5 happens to recognize via two different bound backbone shapes. We use "register" only as shorthand for "which of the two backbone conformations," never to imply a shared sequence.

A central methodological problem in evaluating this kind of claim is that "resembles the native peptide" has historically been operationalized as Cα RMSD to a single crystal structure, an unscaled quantity: a design at 1.2Å cannot be judged "inside" or "outside" a conformational envelope without knowing how wide that envelope is. We address this by calibrating acceptance thresholds against a native 370K/50ns MD ensemble of both conformations (Methods §2.4), and by requiring three independent criteria to agree (proximity, anchor identity, burial depth) rather than relying on proximity alone, which is separately confounded by a backbone-threading artifact (§3.4) that a raw-RMSD-only criterion cannot detect.

We report five results, ordered to lead with the central empirical finding (recovery and crossing both occur and are quantifiable) before addressing what gates their occurrence, what does not explain their rate, a methodological artifact that must be corrected for, and the underlying mechanism.

2 Methods

2.1 Structural system and conformational-state definition

The system is HLA-A*02:01 (α1/α2/α3, chain A) with β2-microglobulin (chain B), a 10-residue peptide (chain C), and the DMF5 TCR α/β V-domains (chains D/E), crystallized in two forms: 6AM5, bearing the peptide SMLGIGIVPV ("GIG"), and 6AMU, bearing the peptide MMWDRGLGMM ("DRG") — two different decamer sequences, not one sequence in two groove registers. Component-wise Cα RMSD between the two crystals (computed after superposing the rigid MHC α1/α2 platform) localizes essentially the entire 2.87Å difference between the two complexes to the peptide backbone; the MHC groove floor (0.20–0.39Å), CDR3α (0.59Å), and CDR3β (0.47Å) are rigid between the two forms. "Register" is used only as a name for a peptide's bound Cα conformation in a common reference frame (§2.3); it carries no sequence-identity meaning here.

2.2 Backbone generation

De novo peptide backbones were generated with RFdiffusion (`Complex_base_ckpt.pt`, SE3nv environment), contig specification `A1-180/0 B1-100/0 D1-115/0 E1-120/0 10-10` (MHC/B2M/TCRα/ TCRβ held fixed as structural context; the terminal `10-10` block is fully de novo with no peptide-backbone residues templated), `diffuser.T=30` (full reverse-diffusion trajectory) throughout. Conditioning was varied solely via `ppi.hotspot_res`, ranging from 3 to 42 receptor-side residues across 55 (crystal $\times$ conditioning) cells drawn from 5 campaigns (internal names: `grind`, `ladder`, `promising`, `denovo30`, `maxcond`), totaling 1,295 scoreable 10-residue designs after filtering (§2.3). We name each conditioning cell by its hotspot count (e.g. $\texttt{C18} = 18$ receptor-side hotspot residues) rather than by internal campaign codename, since contact count is the property under discussion throughout §3; where the same count was tested on both crystals we disambiguate with a crystal suffix, e.g. `C24 (6AM5)` vs. `C24 (6AMU)`. This is the corpus used for the register test in §3.1–3.4; it is distinct from (i) the motif-templating ladder (§3.5), a separate small set of cells (`fixall`$\rightarrow$`fix0`, `rfd_recover` campaign, $N = 10$ designs/crystal/level) that still carry the full rich hotspot list but vary how much of the native peptide backbone is additionally supplied as a template, and (ii) the null-baseline campaign (§3.6), a new set of cells (`null0`) carrying *no* hotspots and no template at all. Of the 55 conditioning cells, 16 pool draws from more than one campaign launched at different times under the same (crystal, conditioning) specification — e.g., the `C3`–`C12` contact-count ladder draws from `denovo30`, `ladder`, and `promising` jointly, and `mhc`/`mhc_tcr2`/`tcr2` draw from `denovo30` and `grind` jointly. We disclose this because it means the 55 cells are not 55 single-shot, mutually independent draws; they are 55 conditioning *specifications*, several of which were incrementally added to across sequential campaign launches. This is consistent with — and does not undermine — the §3.3 finding that per-cell hit count tracks cumulative draw count, since draws accumulated for the same cell over time are exactly the repeated-sampling process that finding describes.

2.3 Register scoring

For each design, the MHC α1/α2 Cα trace is superposed onto a common reference frame via robust (outlier-trimmed) Kabsch alignment; the peptide Cα coordinates are transformed into that same frame, making them directly comparable to the two native reference conformations (GIG, DRG; each a fixed (10,3) Cα array in the same frame) and to a native MD trajectory processed identically.

A design is scored on three criteria, each thresholded against a native MD ensemble of the *target* conformation (own conformation for "recovery," alternate conformation for "crossing"):

1. **Proximity** — mean Cα RMSD to the target's 10-residue reference $\leq$ (native MD mean $+ 3\sigma$) for that conformation.
2. **Anchor identity** — the design's Cα position with minimum distance to a fixed F-pocket centroid (defined from six HLA-A*02 F-pocket residues) must equal the target's native anchor position (P10 for GIG, P9 for DRG).
3. **Burial depth** — that minimum F-pocket distance must fall within (native MD mean $\pm 3\sigma$) for the target conformation.

All three must hold simultaneously. This retires two previously used but unreliable proxies: raw RMSD-only thresholding (which never registers a crossing under any threshold in the 1.4–2.5Å range tested historically, a false-negative failure mode) and F-pocket-argmin proximity alone without a depth requirement — 388/1,295 designs (30%) satisfy this alone, which we consider false-positive-prone, since it is spoofable by centroid-adjacent scatter unrelated to genuine anchor burial.

Threading classification. Each design's F-pocket-proximal anchor position was additionally classified as *forward* (positions P7–P10 occupy the F-pocket, matching both native conformations' actual anchor placement) or *reverse* (P1–P4), with P5–P6 as an ambiguous buffer zone; this classification was derived from the empirical bimodal distribution of anchor position across the full corpus (§3.4) and is applied as a filter before all recovery/crossing statistics reported in §3.1–3.3 (forward-threaded subset, $N = 600$).

2.4 Native MD calibration

Explicit-solvent MD (Amber ff19SB force field, TIP3P water, 12Å periodic box, 0.15M NaCl, 2fs timestep, minimization to convergence, $\leq$10,000 steps) was run for each crystal structure at 370K (2ns equilibration, 50ns production; Tamarind job identifiers `lbv69` for 6AM5 and `p1yv2` for 6AMU). A convergence diagnostic found both trajectories still drifting over the production run (net $+0.25$Å for 6AM5, $+0.82$Å for 6AMU) with autocorrelation times of 5.3ns (6AM5) and 10.7ns (6AMU), giving effective independent sample counts of approximately 9 and 5 (of 1,000 raw frames) respectively; acceptance thresholds reported

conformation	proximity	F-pocket depth
GIG (6AM5)	$\leq$1.49Å	[3.50, 7.48]Å
DRG (6AMU)	$\leq$2.10Å	[3.62, 7.08]Å

Table 1: Calibrated acceptance bands (370K/50ns MD).

here should be treated as current-best estimates, not converged values (see Limitations, §5). The resulting bands are:

The two native conformations remain cleanly separated basins (GIG frames are never closer than 2.40–2.73Å to DRG's reference, and vice versa), confirming the 2.87Å crystal-to-crystal difference reflects genuine thermal separation rather than calibration noise.

2.5 Null/background campaign

To measure the chance rate at which RFdiffusion produces a backbone landing inside either acceptance envelope with **no** conditioning information at all, we generated a separate set of designs (`null0` cells, one per crystal) using the same contig (`10-10`, no template) but omitting `ppi.hotspot_res` entirely — the receptor chains are still held fixed as structural context, but no contact information is supplied. This is distinct from the `fix0` cells used in the templating ladder (§3.5), which — despite the name — still carry the full rich hotspot list; `fix0` varies template fraction at fixed (maximal) hotspot conditioning, whereas `null0` removes hotspot conditioning entirely. Status and current N are reported in §3.6.

2.6 Statistical analysis

Proportions are reported with Clopper–Pearson exact 95% confidence intervals. Comparisons between conditioning tiers use Fisher's exact test; where one cell of the 2×2 table is zero, we additionally report a Haldane–Anscombe continuity-corrected odds ratio (+0.5 to all cells) with its associated 95% CI. Where multiple (55) conditioning cells were compared to identify an outlier cell, the resulting p-value is Bonferroni-corrected for 55 comparisons. The relationship between per-cell sample size and per-cell hit count was assessed by Pearson correlation across all 55 cells.

3 Results

We generated 1,295 de novo peptide backbones across 55 conditioning cells and scored each against the three-criterion register test described in §2.3. Two designs stood out immediately: one closely matched its own native conformation, and another matched the *alternate* conformation better than its own. Reaching a clean count of these events, however, first required resolving a structural ambiguity present in every conditioning scheme tested — described in §3.4 — which we identified during this analysis and corrected for before computing any of the statistics below. We present the headline result first (§3.1), then work outward to what explains its pattern (§3.2–3.3), the correction that made it trustworthy (§3.4), and the mechanism underneath (§3.5), closing with the design and status of a direct null-baseline measurement (§3.6).

outcome	count/N	rate (95% CI)
Own recovery, full pool	1/1,295	0.077% (0.002–0.43%)
Alt. crossing, full pool	4/1,295	0.309% (0.084–0.79%)
Own recovery, fwd. subset	1/600	0.167% (0.004–0.93%)
Alt. crossing, fwd. subset	4/600	0.667% (0.18–1.70%)

Table 2: Recovery and crossing rates, full pool ($N = 1{,}295$) and forward-threaded subset ($N = 600$; §2.3, §3.4), with Clopper–Pearson exact 95% CIs.

residue	contacts	residue	contacts
A167	10	A70	6
E97 (TCRβ)	11	A63	6
A147	9	A67	5
A77	8	A80	5
A159	8	A97	5
A99	8	A143	5
A66	8	D30 (TCRα)	7
A155	7	E96 (TCRβ)	4

Table 3: Peptide-proximal heavy-atom contact counts ($\leq$5Å), 6AM5 crystal structure.

3.1 Both recovery and crossing occur at low, bounded rates

All three criteria (proximity, anchor identity, depth) were satisfied simultaneously by each reported design; none is a marginal call on a single axis. The closest crossing design (Cα RMSD 1.85Å to the alternate conformation's mean, correct anchor position, correct depth) is 1.26Å from a specific frame sampled at 41.4ns into that conformation's MD trajectory — closer to an observed thermal conformation than to the register's time-averaged structure.

3.2 Conditioning breadth, not composition, gates the outcome

Having established that recovery and crossing occur at all, the natural next question was whether the five qualifying designs were spread evenly across the 55 conditioning cells or concentrated in a particular kind of cell. Conditioning hotspot sets ranged from 3 residues (sparse) to 37–42 (rich). We evaluated whether the composition of "rich" cells reflected anything beyond generic contact frequency by computing, directly from the 6AM5 crystal structure, the number of peptide-proximal heavy-atom contacts ($\leq$5Å) per receptor residue:

Every hotspot used in the richer conditioning cells is among the top-ranked residues by this generic, distance-based metric; no residue included at the rich end reflects a curated or mechanistically distinct selection beyond higher raw contact count on both the MHC and TCR side. We therefore treat "rich" as "broad coverage of high-contact-degree residues" rather than as a biologically distinguished category.

Under this operationalization: designs from cells covering $\geq$9 high-contact-degree residues produced 5/718 hits (0.70%, 95% CI 0.23–1.62%); cells limited to the top 3–5

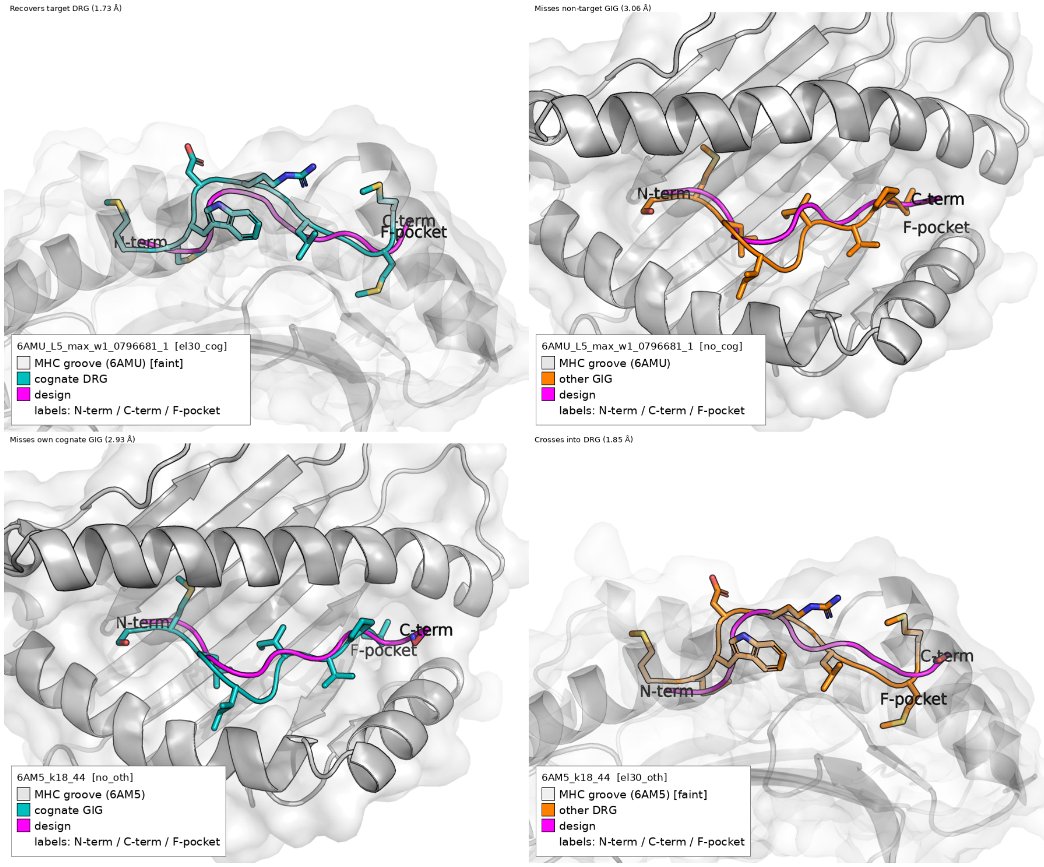

Figure 2: **Recovery and crossing are register-specific, not generic proximity.** Each row is one design shown against both conformations, establishing that the match to its reported conformation is not simply "close to either." Top row: the recovering design (magenta) is 1.73Å from its own target DRG (teal, left) but 3.06Å from the non-target GIG (orange, right) — nearly 2× farther. Bottom row: the closest crossing design (magenta) is 2.93Å from its own cognate GIG (teal, left) — the conformation it was *not* trying to match — but 1.85Å from the DRG conformation it crossed into (orange, right; 1.26Å to the single closest individual MD frame, discussed below). Both designs sit substantially closer to the conformation they are credited with matching than to the alternative.

residues produced 0/587 (95% CI 0–0.63%). Fisher's exact test: $p = 0.068$ (Haldane–Anscombe-corrected OR$= 9.06$, 95% CI 0.50–164, reflecting the small event count). This result is suggestive rather than conclusive on its own, but is consistent with conditioning breadth being necessary, even though (§3.3) it is not sufficient to predict which specific rich design succeeds.

3.3 Above the gate, sampling depth explains hit frequency

Conditioning breadth (§3.2) explains why sparse cells never hit, but it does not by itself explain which *rich* cell produces a hit and which does not — the five qualifying designs came from only 3 of the many rich cells tested. To ask whether some rich conditioning schemes are simply better than others, we ran a fixed, matched campaign: 26 designs were generated under each of 8 different rich conditioning schemes (5–42 contacts, both crystals), all drawn before any cross-scheme comparison was made, so that no scheme could be favored by having received more attempts than another:

conditioning	contacts	best-of-26 (Å)
C9	9	2.06
C18	18	2.06
C5	5	2.24
C42 (6AMU)	42	2.27
C37 (6AM5)	37	2.28
C24 (6AM5)	24	2.40
C18 (6AMU)	18	2.42
C24 (6AMU)	24	2.55

Table 4: Best-of-26 distance to target across 8 matched rich-conditioning cells.

All eight cells converge to a 0.49Å-wide band regardless of a nearly 9-fold range in contact count. Extending the single best-performing cell (C9) from 26 to 133 draws did not improve on the original best (2.06Å unchanged); the 107

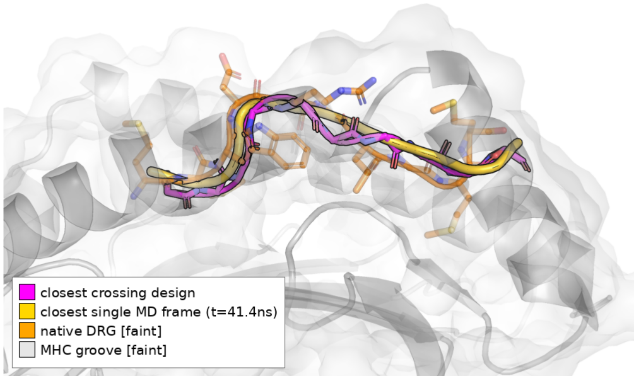

Figure 3: **The closest crossing design against the single closest real MD snapshot.** Magenta: the closest crossing design (`C18_44`). Gold: the individual 370K/50ns MD frame it is nearest to ($t = 41.4$ns into the DRG trajectory, 1.26Å away) — an actual sampled thermal conformation, not the register's time-averaged mean (orange, faint, shown for reference at 1.85Å). The two traces nearly coincide.

additional draws contributed a worse best (2.36Å) and substantially worse median (13.7Å vs. 4.1Å for the first 26) — consistent with resampling a fixed distribution rather than converging toward the target.

Across all 55 conditioning cells (not just the 8 in the controlled comparison), per-cell hit count correlated with per-cell draw count (Pearson $r = 0.657$, $p < 0.0001$). The single cell with the highest raw hit count (C18, combined across both crystals: $3/186 = 1.61\%$, vs. $2/1{,}119 = 0.18\%$ for all other cells combined; Fisher's exact $p = 0.023$) does not survive Bonferroni correction across the 55 cells examined (corrected $p = 1.0$). We conclude that, conditional on clearing the conditioning-breadth gate identified in §3.2, the identity of the conditioning scheme does not detectably predict outcome; the number of designs drawn does.

A note on directional asymmetry. The hit set contains 4 crossings and 1 recovery — the opposite of what a hypothesis of "conditioning biases toward its own target" would predict. With $n = 5$ total events, a two-sided exact binomial test cannot distinguish this split from 50/50 ($p = 0.375$), so we do not read this as evidence of a directional bias one way or the other.

3.4 A threading artifact affects half of backbones and must be filtered

The results in §3.1–3.3 are already reported net of a correction we describe here. Early in this analysis, per-conditioning distributions of Cα RMSD were consistently bimodal — a near-native mode and a second, far mode roughly matched in size — under every conditioning scheme inspected, regardless of contact count. Both native conformations anchor a C-terminal-half residue (P9 or P10) in the MHC F-pocket; tracing the far mode to its cause showed that, in the absence of any explicit directional constraint, de novo backbones instead anchor their N-terminal residue (P1–P4) in that same pocket in close to half of all draws — a mirror-image backbone, not a genuine register variant, which raw Cα RMSD to a crystal structure cannot distinguish from a poorly-folded forward-threaded design. This was quantified in two representative conditioning cells:

Having already found in §3.2–3.3 that conditioning breadth and identity shape whether recovery and crossing occur, we checked directly whether they also shape which way the backbone threads. Across all 36 conditioning cells with a known hotspot count (3–37 contacts, $N = 1{,}160$ designs), the forward-threaded fraction showed no relationship to contact count (Pearson $r = 0.154$, $p = 0.37$; Spearman $\rho = -0.019$, $p = 0.91$) and stayed within a 40–58% band at every contact-count tier tested. Threading direction is therefore independent of conditioning entirely, not merely similar between the two cells shown above. Per-position deviation profiles for the forward and reverse sub-populations are approximately mirror images of one another; pooling them without this classification produces a spurious "elevated deviation at both termini" signature that is an artifact of population mixing rather than a genuine failure mode. All recovery/crossing statistics in §3.1–3.3 are reported on the forward-threaded subset ($N = 600$) for this reason.

3.5 Templating, not contacts, controls a localized register signal

Sections 3.2–3.4 establish *when* recovery and crossing occur (broad conditioning, §3.2), and what does not further explain their rate (which specific rich scheme is used, §3.3, or backbone threading, §3.4) — but not *why* contact conditioning plateaus at all. To test this directly we ran a separate experiment: instead of conditioning on receptor-side contacts, we fixed an explicit fraction of the native peptide backbone itself as a structural template and let RFdiffusion generate only the remainder. Register recovery scaled sharply with the fraction of geometry supplied this way:

Templating 8/10 residues recovers register to sub-thermal precision (0.39Å, tighter than the native peptide's own $\sim$1.0Å thermal envelope at 370K). By contrast, contact-hotspot count alone — up to 42 residues, exceeding the information content of the 8-residue geometric case by residue count — does not break a $\sim$2–2.5Å floor, and hotspot count does not correlate with recovery quality (Spearman $\rho \approx 0.17$, not significant; note the `fix0` level of this same ladder, which retains the *full* rich hotspot list but zero template, is the worst-performing level at $\sim$11.4Å median — direct evidence that hotspot conditioning by itself, at its richest, does not recover the target conformation). We interpret this as reflecting distinct mechanisms of action: a contact hotspot biases the predicted position of a *neighboring*, receptor-facing residue, but does not directly constrain a peptide residue's own backbone dihedral angles — the coordinate on which register is actually encoded.

This localization is direct and measurable: per-residue deviation between the two native conformations, expressed in native-MD thermal-fluctuation units (z = inter-

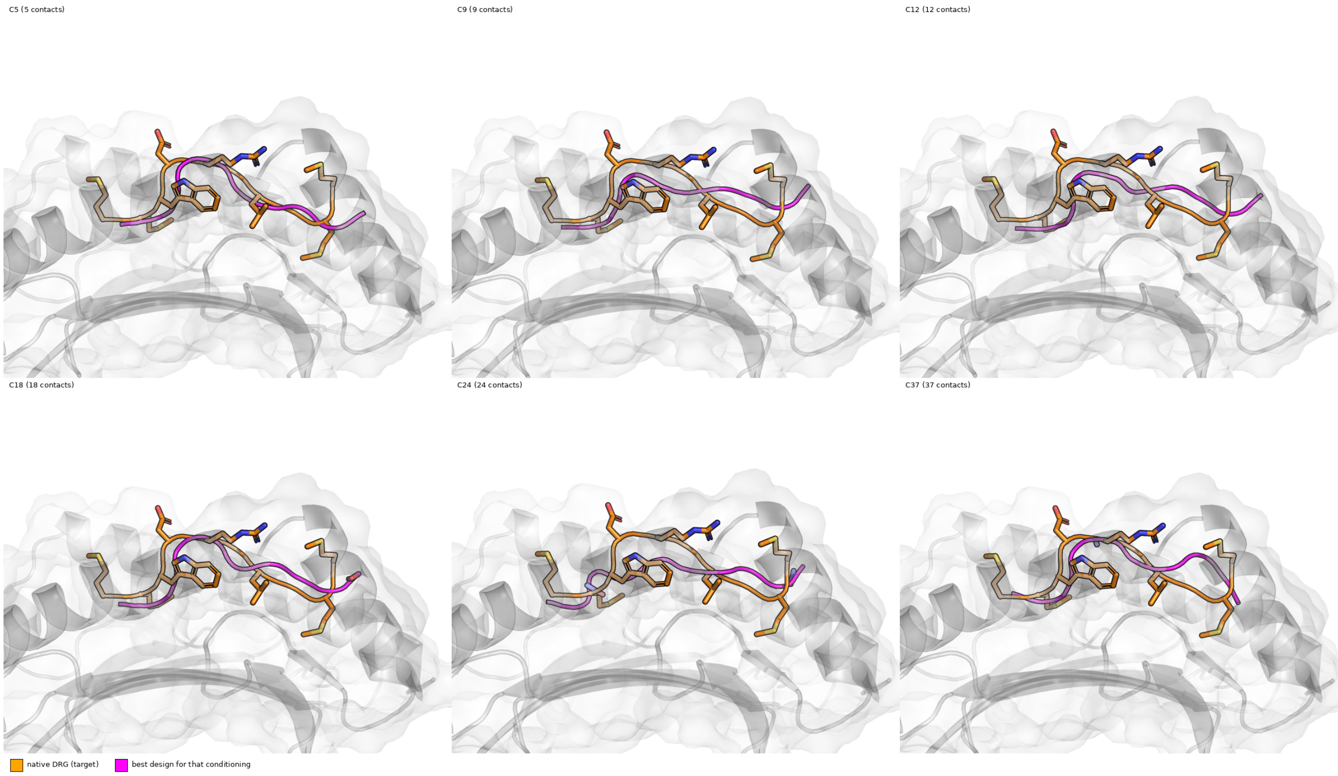

Figure 4: **Best design per rich conditioning, one panel each.** The best (closest-to-target) design (magenta) found for each of 6 rich conditioning cells (5–37 contacts), each overlaid on the native DRG target (orange) against the MHC groove (white surface). All six land at a similar distance from the target regardless of contact count — the structural counterpart of the 0.49Å-wide best-of-26 band in Table 4. Restricted to the rich tier; combining with sparse-tier cells here would conflate the §3.2 and §3.3 comparisons.

conditioning	contacts	N	forward	reverse	ambiguous
C18	18	186	93 (50.0%)	87 (46.8%)	6 (3.2%)
C9	9	204	97 (47.5%)	92 (45.1%)	15 (7.4%)

Table 5: Threading classification, two representative conditioning cells.

residues templated (of 10)	10	8	6	4	2	0
median Cα RMSD to own conformation (Å)	0.07	**0.39**	1.45	1.94	2.34	$\sim$11.4

Table 6: Register recovery vs. fraction of native peptide backbone directly templated.

conformation separation / native positional RMSF at that position), rises from statistical noise at the N-terminus to a pronounced signal at the C-terminal anchor ($z = 0.6$ at P1; 1.3 at P4; 4.9 at P7; 8.1 at P8; 8.5 at P9; 14.5 at P10), with native MD RMSF at these C-terminal positions remaining tight ($\approx$0.37Å), confirming the signal reflects genuine geometric separation rather than amplified noise. This motivates the anchor-identity-based scoring used throughout (§2.3): a whole-peptide-averaged RMSD dilutes this sharp, spatially confined signal across five register-neutral N-terminal positions.

3.6 Null/background campaign: interim result and status

Sections 3.1–3.3 report a corpus-wide hit count of 5/1,295 across a range of contact-conditioning schemes, but do not by themselves establish how this compares to a fully unconditioned RFdiffusion draw — the comparisons above are between conditioning tiers, not against a true zero-information background. To measure that background directly, we launched a dedicated campaign (`null0`, §2.5) generating de novo backbones with the receptor held fixed as structural context but **no** `ppi.hotspot_res` and no

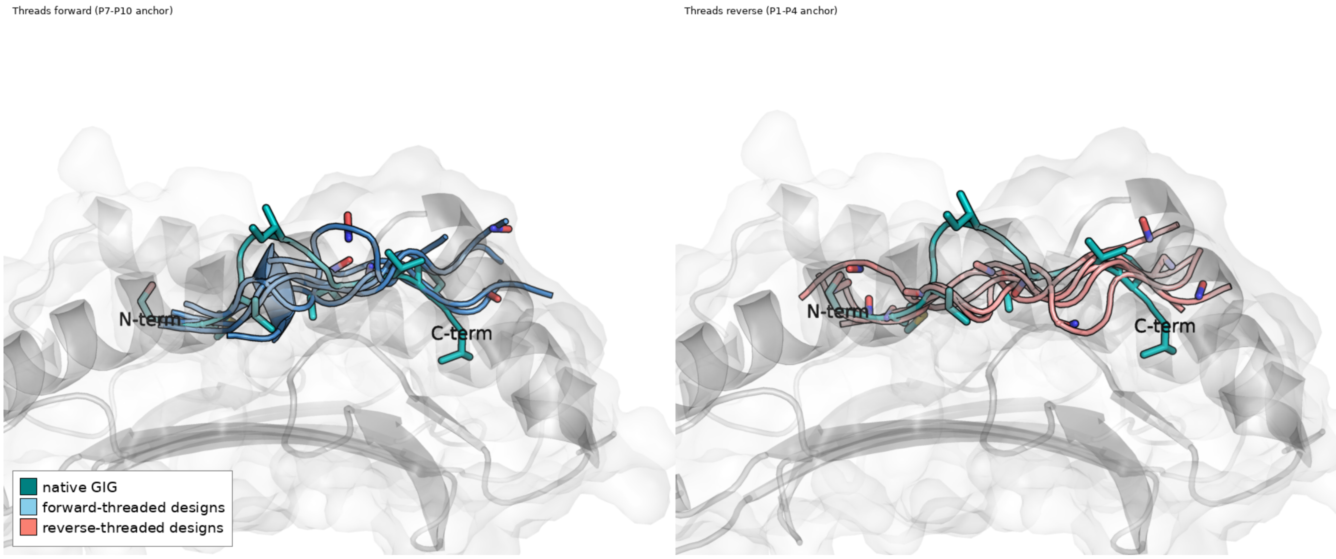

Figure 5: **Forward vs. reverse threading, structural overlay** (C18, 5 representative designs per panel, from a 186-design cohort). Native GIG peptide in teal (N-term/C-term labeled) against the MHC groove (white surface). Left panel: forward-threaded designs (sky blue) run N-to-C the same direction as the native peptide. Right panel: reverse-threaded designs (salmon) anchor the *opposite* end in the same F-pocket — the mirror-image backbone described in §3.4, made directly visible rather than inferred from a distribution.

peptide template, targeting $N = 150$ per crystal (300 total), an order of magnitude larger than any single existing sparse-tier cell.

At an interim checkpoint of $N = 28$ (14/crystal; campaign ongoing toward the full target), **0 of 28** designs satisfy any of the three acceptance criteria, let alone all three jointly. More strikingly, the raw proximity values are not merely on the wrong side of the acceptance threshold — they are an order of magnitude outside it: mean Cα RMSD to the nearer native conformation is 30.9Å (6AM5-context draws) and 17.6Å (6AMU-context draws), against a $\leq$1.49–2.10Å acceptance band and a $\sim$2–2.5Å floor for the *best* contact-conditioned draws anywhere in the main corpus (§3.3). The closest single unconditioned design (3.3Å) is still farther from its native conformation than every one of the 5 accepted hits in §3.1. This interim result (Clopper–Pearson 95% CI for 0/28: 0–12.3%; Fisher's exact vs. the 5/1,295 corpus-wide rate, $p = 1.0$, underpowered at this N to reach significance) is directionally consistent with contact conditioning providing real, if modest, information — even sparse conditioning anchors the peptide near the groove in a way that zero conditioning does not — but is not yet large enough to formally establish the background rate. We will update this section with the completed $N = 300$ comparison once the campaign finishes. We note that the existing `fix0` templating-ladder cells ($N = 5$/crystal, full hotspot list, zero template; §3.5) are *not* a substitute for this measurement, since they still carry the richest hotspot conditioning tested anywhere in this corpus — they are a positive-conditioning control, not a null.

4 Discussion

Three findings jointly define the current capability boundary. First, recovery and crossing of peptide register by contact-conditioned de novo backbone generation are real, structurally validated events, not statistical noise or scoring artifacts — each reported design passes an independent three-criterion test referenced against thermal motion rather than a static target. Second, achieving either outcome appears to require a minimum breadth of conditioning (coverage of high-contact-degree residues on both the MHC and TCR side), but not a specific curated subset of them — the operative variable is coverage, not composition. Third, once that breadth is present, outcome frequency is governed by how many designs are sampled rather than by which qualifying conditioning is chosen — a result that reframes the practical question from "which contacts should be used" to "how much sampling budget is available," a comparatively more tractable lever.

The mechanistic account in §3.5 is consistent with this picture: contact hotspots constrain a *neighboring* residue's predicted position without directly constraining a peptide residue's own backbone torsions, so increasing contact count broadens the region of conformational space the model is nudged toward without narrowing precision at the specific coordinate (anchor-residue backbone geometry) that defines register. Direct geometric templating, which supplies that coordinate explicitly, achieves sub-thermal precision at 8/10 residues — an upper bound on what the architecture can represent, distinct from what contact-only conditioning can currently retrieve.

The N→C threading artifact (§3.4) — that roughly half of all unconstrained backbones bury the mirror-image terminus in the anchor pocket, invisible to raw Cα RMSD — is a genuine, transferable caution for anyone scoring de novo peptide backbones by RMSD alone, independent of the specific recovery/crossing rates reported here. The null-baseline experiment now underway (§3.6) is intended to establish how the 5/1,295 corpus-wide rate compares to a true zero-information background, which the tier comparisons in §3.2–3.3 cannot by themselves resolve.

5 Limitations and scope

- **Backbone (Cα) scope only.** No sequence-, side-chain-, or rotamer-level claim is made; RFdiffusion's output at this stage carries no designed sequence, and any downstream ProteinMPNN sequence-design results are a separate body of work not represented here. This analysis deliberately stops at backbone generation and does not run ProteinMPNN.
- **Calibration convergence.** The 370K acceptance thresholds (§2.4) derive from a single 50ns MD trajectory per register that had not fully converged at the time of analysis (autocorrelation times of 5.3–10.7ns imply effective sample sizes of $\sim$5–9, not the nominal 1,000 raw frames). We expect further simulation to refine these thresholds rather than overturn the qualitative result, given the margin by which the reported designs clear them, but exact Å cutoffs should be treated as current-best rather than final.
- **No completed null/background baseline at the time of this writing.** §3.6 describes a campaign launched to measure this directly; until it completes, the 5/1,295 corpus-wide rate should be read as "observed hit rate across the conditioning schemes tested," not as "rate in excess of what unconditioned generation would also produce," since those are not yet known to be different numbers.
- **Overlapping campaign membership.** 16 of the 55 conditioning cells pool draws from more than one campaign launched at different times (§2.2); this is disclosed for transparency and is consistent with, not a threat to, the draw-count correlation reported in §3.3.
- **Small absolute event counts.** All corpus-wide rates in §3.1–3.2 rest on 5 total qualifying designs (1 recovery + 4 crossings) out of 1,295; the rich-vs-sparse comparison (§3.2) and the C18-specific comparison (§3.3) both carry correspondingly wide confidence intervals and should be read as suggestive rather than conclusive pending a larger, purpose-built confirmatory sample.
- **Single system.** All results pertain to one TCR–pMHC pair, one class-I allele, and one force field; no claim is made regarding generalization to other cross-reactive systems, TCRs, or HLA alleles.

6 Data and code availability

Design generation, scoring, and MD analysis scripts are maintained in the project repository (`py/score_denovo_designs.py` for register scoring and threading classification; `py/native_md_components.py` for MD trajectory processing; `jobs/` for RFdiffusion campaign drivers, including `jobs/run_null_baseline.sh` for the §3.6 null/background campaign). Corresponding analysis notebooks (numbered `00–16`, grouped by methodology) are indexed in `ANALYSIS_WALKTHROUGH.md`; the MD-calibrated baseline and joint acceptance test underlying §2.3–2.4 and §3.1 are implemented in `01_md_calibrated_baseline.ipynb`.